Music Classification and Statistical Testing

AI in Music, Winter Term 2025/2026

20.11.2025

Exercise 1 [Genre Classification with Decision Trees, 4 points]

In this exercise, you will train decision trees for binary genre classification between pop and rock music. For this, you can use the sklearn implementation of `sklearn.DecisionTreeClassifier`. Use the features and labels contained in the files `PopRock-TS120-FP1.arff` as training set, and `PopRock-TAS120-FP1.arff` as test set. Note that the final category in the arff files is the label indicator, where `AG_Pop` corresponds to pop music, and `NOT_AG_Pop` corresponds to rock music.
Conduct the following experiment: Create 10 decision trees with 10% of the training set, randomly chosen each time. Then, create 10 more decision trees using 50% of the training data.
Use the test set to measure the difference between the mean precision, recall and F1-scores of trees created with more and less data.

Exercise 2 [Statistical Testing, 4 points]

Test whether the precision, recall and F1-scores of your decision trees are normally distributed. Then, use non-parametric statistical testing to see if the differences in performance are statistically significant ($\alpha = 0.05$).

Exercise 3 [Feature Evaluation, 2 points]

Visualise one of your trees for the low-data case and one for the large-data case, using the tutorial given by sklearn.
Finally, investigate the features your decision trees have chosen to classify the data. Which features are chosen most commonly?